"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh disciplane issues!"

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4.)

$$\begin{vmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -3\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{vmatrix}$$

$$\begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -11 & 1 & 0 \\ 0 & 6 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{vmatrix} \begin{vmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -\frac{1}{2} & 1 \end{vmatrix}$$

a.) Find the unknown enteries α, γ, β, t and m =>

b.) " " eliminated upper triangle matrix =>

c.) Determine the remaining elements

$$\begin{vmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -3\gamma & 1 \\ 2 & \beta & 3 & -\alpha^2 \\ 1 & 14 & -6 & 2 \end{vmatrix} \begin{matrix} -3\alpha R_1 + R_2 = R_2 \\ -2R_1 + R_3 = R_3 \\ -R_1 + R_4 = R_4 \end{matrix} \begin{vmatrix} 1 & \alpha & -\gamma & \alpha \\ 0 & (5-3\alpha^2) & -3\gamma + 3\alpha\gamma & 1-3\alpha^2 \\ 0 & \beta - 2\alpha & 3+2\gamma & \alpha^2 - 2\alpha \\ 0 & 14-\alpha & \gamma - 6 & 2-\alpha \end{vmatrix}$$

$+ \frac{1}{(5-3\alpha)^2} \cdot R_2 \to R_2 \to$ new row 2 = $\quad * \frac{3+3\alpha\gamma}{(5-3\alpha)^2} \quad \frac{1-3\alpha^2}{(5-3\alpha)^2}$

$$(\beta - 2\alpha) R_2 + R_3 = R_3$$

$$(14-\alpha) R_2 + R_4 = R_4$$

$$\alpha \cdot R_2 + R_1 = R_1$$

$$\begin{matrix} 1 & 0 & \left(\frac{3+3\alpha\gamma}{(5-3\alpha)^2}\right)\alpha - \gamma \\ 0 & 1 & -\frac{3+3\alpha\gamma}{(5-3\alpha)^2} \\ 0 & 0 & \\ 0 & 0 & \end{matrix} \to \frac{-3+3\alpha}{(5-3\alpha)^2} \quad (\beta - 2\alpha)_{+\downarrow}$$

$$\to \left(\frac{-3+3\alpha\gamma}{(5-3\alpha)^2}\right) \cdot (14-\alpha) \quad \left(\frac{3+2\gamma}{\gamma - 6}\right)$$

$$\frac{1-3\alpha^2}{(5-3\alpha)^2} \quad \frac{1-3\alpha^2}{(5-3\alpha)^2} \Rightarrow$$

$$\frac{1-3\alpha^2}{(5-3\alpha)^2} \beta - 2\alpha \quad \Rightarrow (14+\alpha) + 2 - \alpha$$

$$+ \alpha^2 - 2\alpha$$

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2.)

a.) its intersection is to occurs 3D (Vol), I can say that the minimum dimension of space is 3D.

b.) if the unknown geometry is 4D we can say it, compose 3D shape if the unknown geometry is 2D we can say it is composed 1D we have (unknown dimension - 1) geometrical shape.

c.) $x+4y-z=6$

$$\begin{bmatrix} 1 & 4 & -1 \end{bmatrix} \cdot \begin{bmatrix} x \\ 1 \\ 2 \end{bmatrix} = 6$$

P F F

Pivot Free

intersection, we should look the nullspace

xy plane $z=0$

$$\begin{bmatrix} 1 & 4 & -1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

P F P

$z=0$ then $x+4=0$

$y=1$ $x=-4$

$$\propto \begin{bmatrix} -4 \\ 1 \\ 0 \end{bmatrix}$$

$N(A)$

and that is 2D inside 3D

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3.) a.) $H^2 = I$ $\quad H = H^T$

$H^2 = (I - 2uu^T)^2 = I^2 - 4uu^T + 4uuu^Tuu^T$

since $u^Tu = 1$

$\hookrightarrow H^2 = I$. $\quad H = H^T$ $H = I$

H is an orthogonal matrix.

b.) $Hu = (I - 2uu^T)u = u - 2uu^T = u - 2u = -u$, $\lambda = -1$.

c.) v orthogonal to u (i.e $u^Tv = 0$) $\rightarrow Hv = (I - 2uu^T)v =$

$v - 2uu^Tv = v \Rightarrow \lambda = 1$

(n-1) orthonormal eigenvectors have a basis, by adding the eigenvector u with this, it is found that H is diagonalizable

d.) uu^T is u_i^2 , $Hu = 1 - 2u_i^2$

Summing this together gives $\sum_{i=1}^{n} H_{ii} = n - 2\sum_{i=1}^{n} u_i^2 = n - 2$

Collecting up the eigenvalues $\sum \lambda_i = (1) - 1 + (n-1)(1) = n - 2$

H gives 1